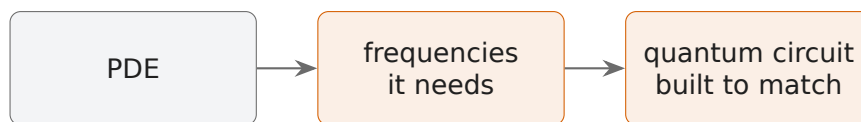


Qapiino

Designing quantum circuits from
the physics they have to solve

Spectrum-Matched Quantum-Assisted Physics-Informed Neural Networks

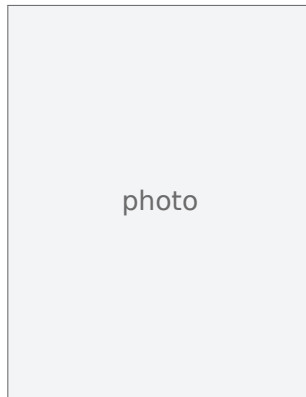


WISER Summer Program 2026 • Industry Challenge (BQP) • Top 10

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About the author



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Role. [One or two lines on what you did in the project.]

Abstract

Physics-informed neural networks learn smooth, slow variations quickly and sharp, fast ones slowly. A quantum circuit that re-uploads its input can only produce a fixed, predictable set of frequencies. We turned that into a design rule, SMCD: read the frequencies a PDE's solution needs from the equation, then build the circuit whose frequency set covers them, without architecture search.

We tested the rule the strict way. Twelve predictions were committed to git before the first experiment, and a script, not us, decided every verdict. Over 191 training runs on six PDEs the design rule did what it promised: the circuit produces nothing outside its designed frequencies, and the designed circuit beat the same circuit with random frequencies by 6.5× on the heat equation, on every seed. The original quantum model did not beat the classical baselines, and our explainability instruments show why. We then took the method to a real problem, waterlogging of farmland from leaking irrigation canals. The exact solution gives a planning map; the original circuit failed to learn it. Finally we built an improved circuit from eight copies of the design, locked new predictions in git and tested it on fresh seeds. On the farmland problem it met all five predictions: 0.09% error, 18× better than a classical network of the same size, and the waterlogged area right to within 1 m. On Poisson its error fell from 1.68 to 0.0037, though there a classical model given the same frequencies is still 3× more accurate.

At a glance

6/15

pre-registered checks
confirmed at the full pro-
tocol (3/15 at submission)

6.5×

lower error: designed
circuit vs. random cir-
cuit, heat equation

450×

lower Poisson error for the
improved circuit on fresh
seeds (1.68 → 0.0037)

5/5

locked farmland predictions
met by the improved
circuit (0.09% error)

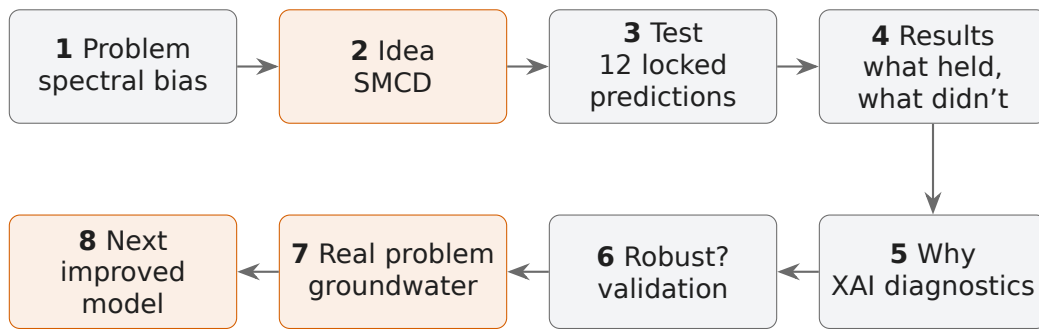
20%

less canal seepage (lining)
dries the whole field

0

thresholds changed
after seeing results

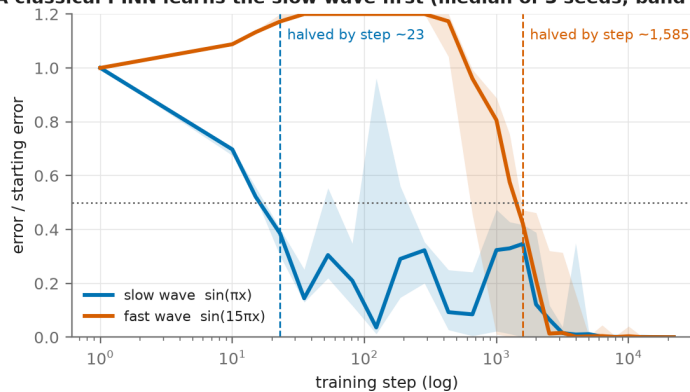
How this report flows



1 The problem: neural networks learn slow waves first

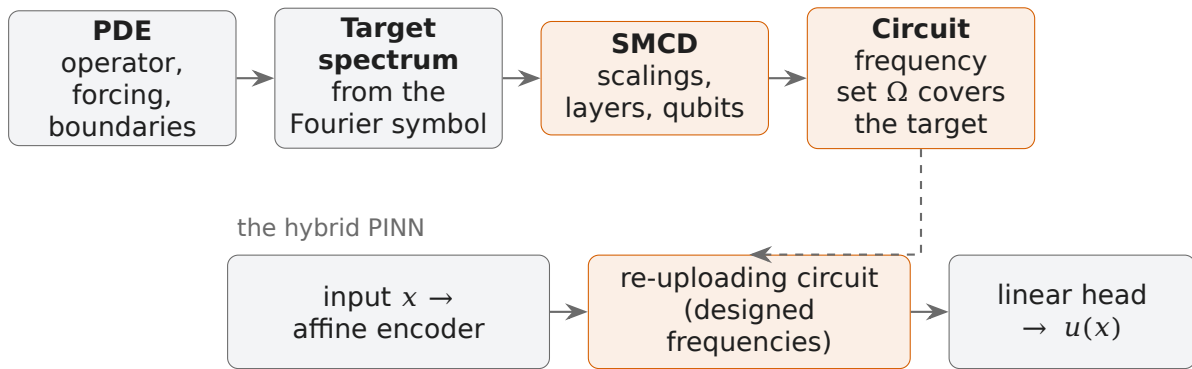
A standard physics-informed neural network (PINN) fits a smooth solution quickly and struggles with fine detail. This is *spectral bias*: the network's learning speed falls with frequency. Below, a classical network halves its error on the slow $\sin \pi x$ part of a Poisson solution within 23 steps, and needs 658–1,585 steps for the fast $\sin 15\pi x$ part.

A classical PINN learns the slow wave first (median of 3 seeds, band = range)

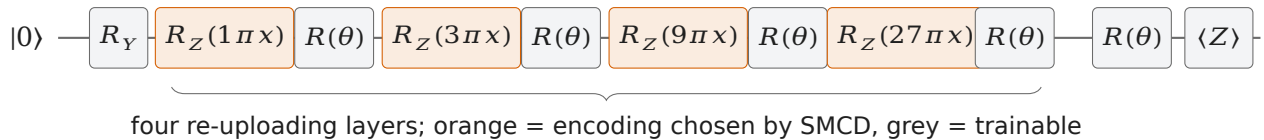


A re-uploading quantum circuit behaves differently. Its output is a Fourier series whose frequencies are fixed by how the input is encoded, so they can be chosen in advance.

2 Our idea: build the circuit from the equation

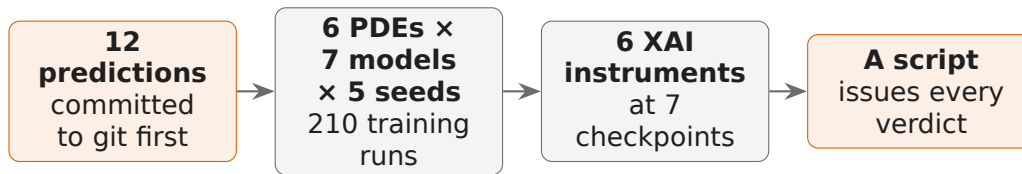


Example. Poisson's solution $\sin \pi x + 0.3 \sin 15\pi x$ needs the frequencies π and 15π . SMCD picks encoding scalings $\pi \cdot \{1, 3, 9, 27\}$ on one qubit with four layers; the circuit can then produce every multiple of π up to 40π , which covers both. No search, and the design is checkable before training.



Encoding scalings	$\pi \cdot \{1, 3, 9, 27\}$
Reachable frequencies Ω	every sum $\pm\pi \pm 3\pi \pm 9\pi \pm 27\pi =$ all multiples of π from -40π to 40π (81)
Needed by the solution	π and 15π (both in Ω)
Trainable parameters	14

3 How we tested it



Problems	Poisson, heat, Burgers, Helmholtz at $k = 4, 10, 20$
Quantum models	q_serial (SMCD design), q_random (same size, random frequencies), q_parallel, q_octave
Classical models	MLP, Fourier features, RFF matched (the same frequencies as the circuit, no quantum)
Instruments	NTK spectrum, per-frequency error, Fisher information, attributions, encoder drift, gradient variance, loss landscape
Rule	a prediction is confirmed only if its pre-set threshold is met and, for comparisons, all five seeds agree

The deadline cut the plan to two seeds. After the challenge we finished it as written: five seeds, both qubit counts for the barren-plateau check, the full training budget. No prediction or threshold was changed.

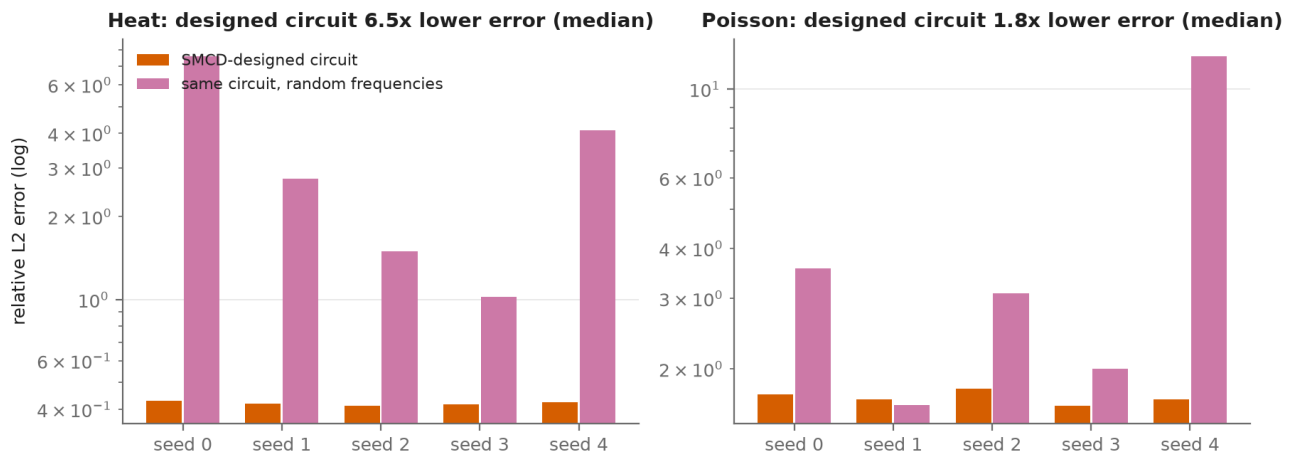
4 Results: what held and what didn't

3/15 confirmed -> 6/15 confirmed

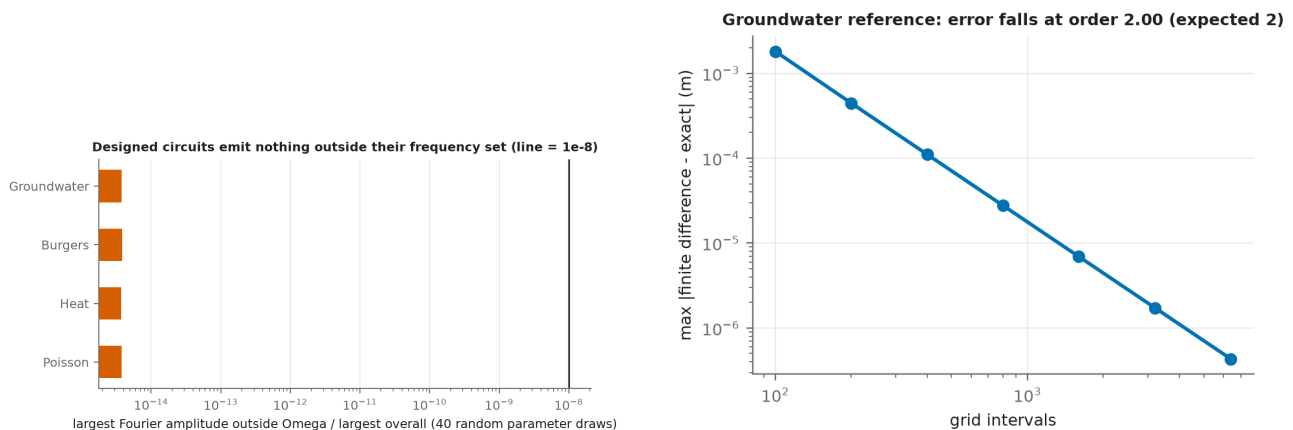
	Submitted, 2 seeds	Full protocol, 5 seeds
PR-1	Refuted	Refuted
PR-2	Refuted	Refuted
PR-3	Refuted	Refuted
PR-4	Refuted	Confirmed
PR-5	Confirmed	Confirmed
PR-6 (poisson)	Inconclusive	Inconclusive
PR-6 (heat)	Inconclusive	Confirmed
PR-7 (poisson)	Insufficient Data	Insufficient Data
PR-8 (poisson)	Confirmed	Confirmed
PR-7 (helmholtz_k10)	Refuted	Refuted
PR-8 (helmholtz_k10)	Refuted	Refuted
PR-9	Refuted	Refuted
PR-10	Insufficient Data	Confirmed
PR-11	Refuted	Refuted
PR-12	Confirmed	Confirmed

- **Held.** The designed circuit beats a random one (heat, 6.5×); its improvement lands on the designed frequencies (99.9% on Poisson); the encoder stays put during training; nothing beats the MLP on the smoothing heat equation, as predicted.
- **Did not hold.** On Poisson the circuit (rel. error 1.68) lost to the MLP (1.08), Fourier features (0.21) and RFF matched (0.33); coverage did not predict error; the NTK did not flatten inside Ω .
- **Careful.** The barren-plateau check passes its rule (slope 0.31), but the fit explains almost nothing ($R^2 = 0.001$): no decay was detected between 4 and 6 qubits.

5 The design rule works

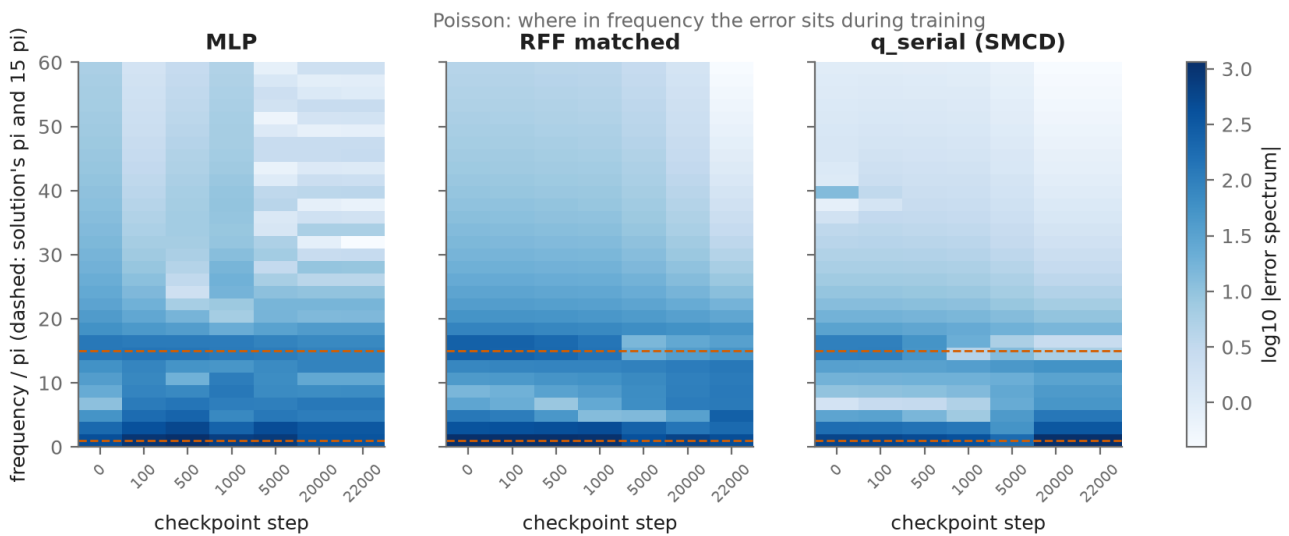


Same circuit, same size, same training: only the frequencies differ. On heat the designed circuit wins on every seed (95% interval 2.5–17.7×). On Poisson it wins on four of five, so the verdict stays inconclusive.

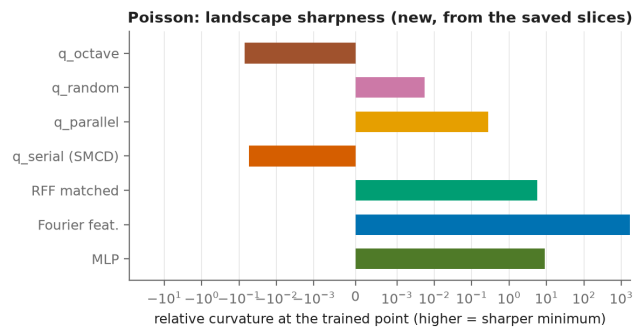
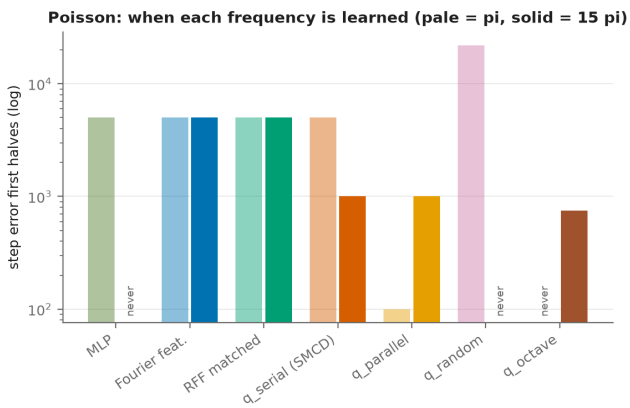


Left: at random parameters, every designed circuit's output outside Ω is at most 4×10^{-15} of its largest component. Right: our reference solver converges at the textbook rate, so the errors we report are the models', not the reference's.

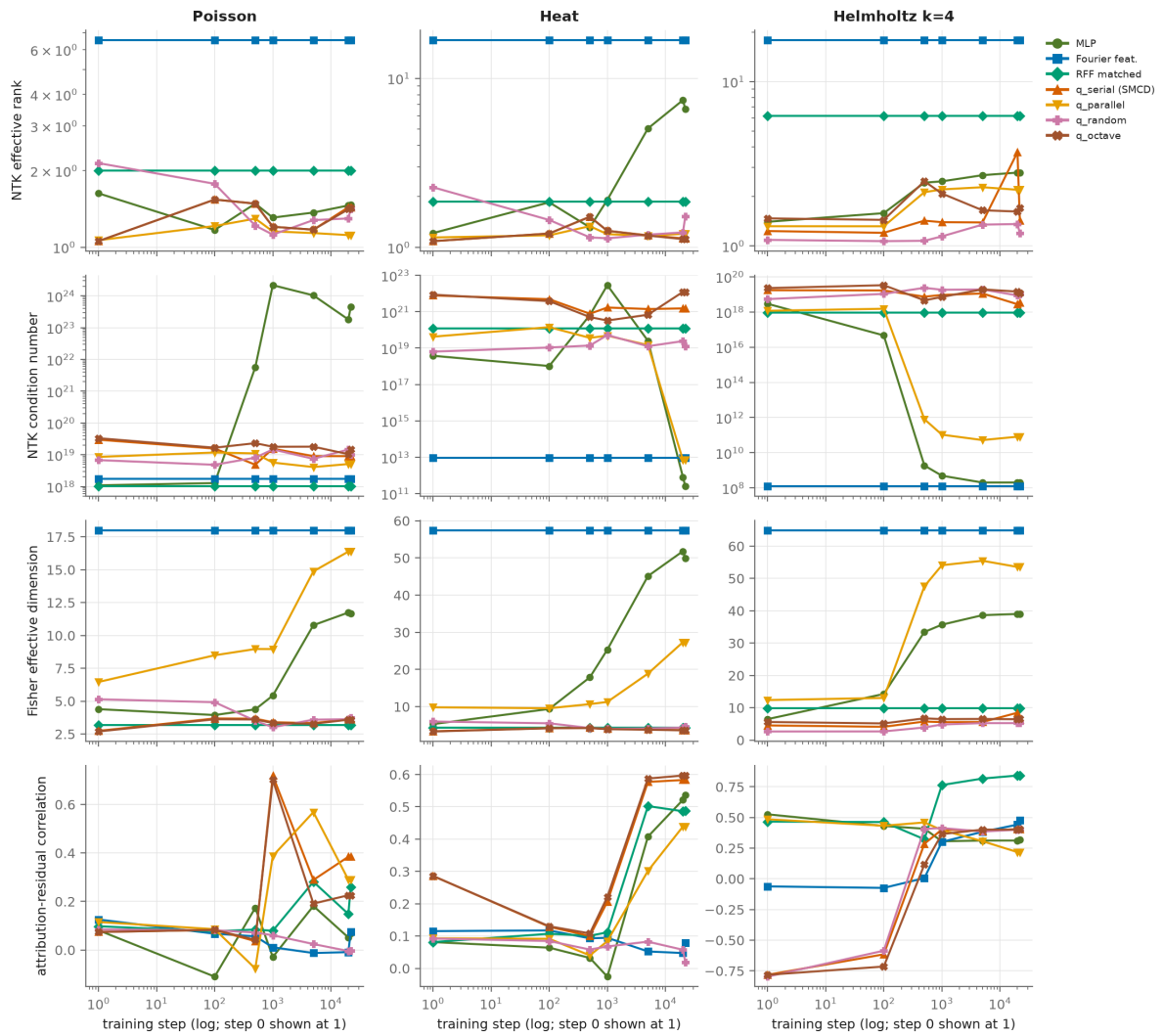
6 Why the circuit loses on Poisson



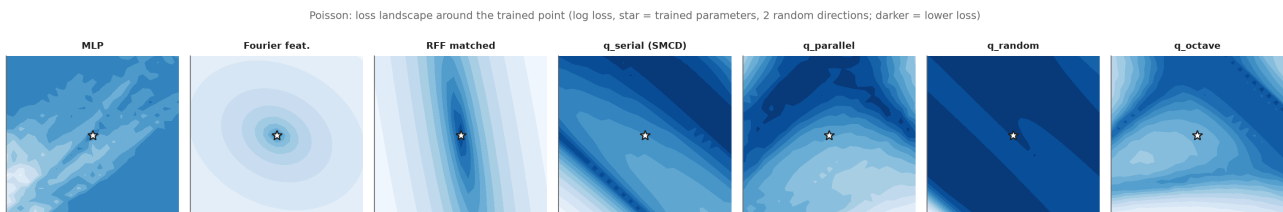
The per-frequency error shows the mechanism. The circuit removes the hard 15π error early (right panel, upper dashed line), then *loses* the easy π component: its π error ends at about twice its starting value on all five seeds. The residual loss weights the 15π mode 225 times more than the π mode, and all of the circuit's wave amplitudes are tied to the same ten angles, so improving one costs the other. The classical model with the same two frequencies sets each amplitude independently and improves both.



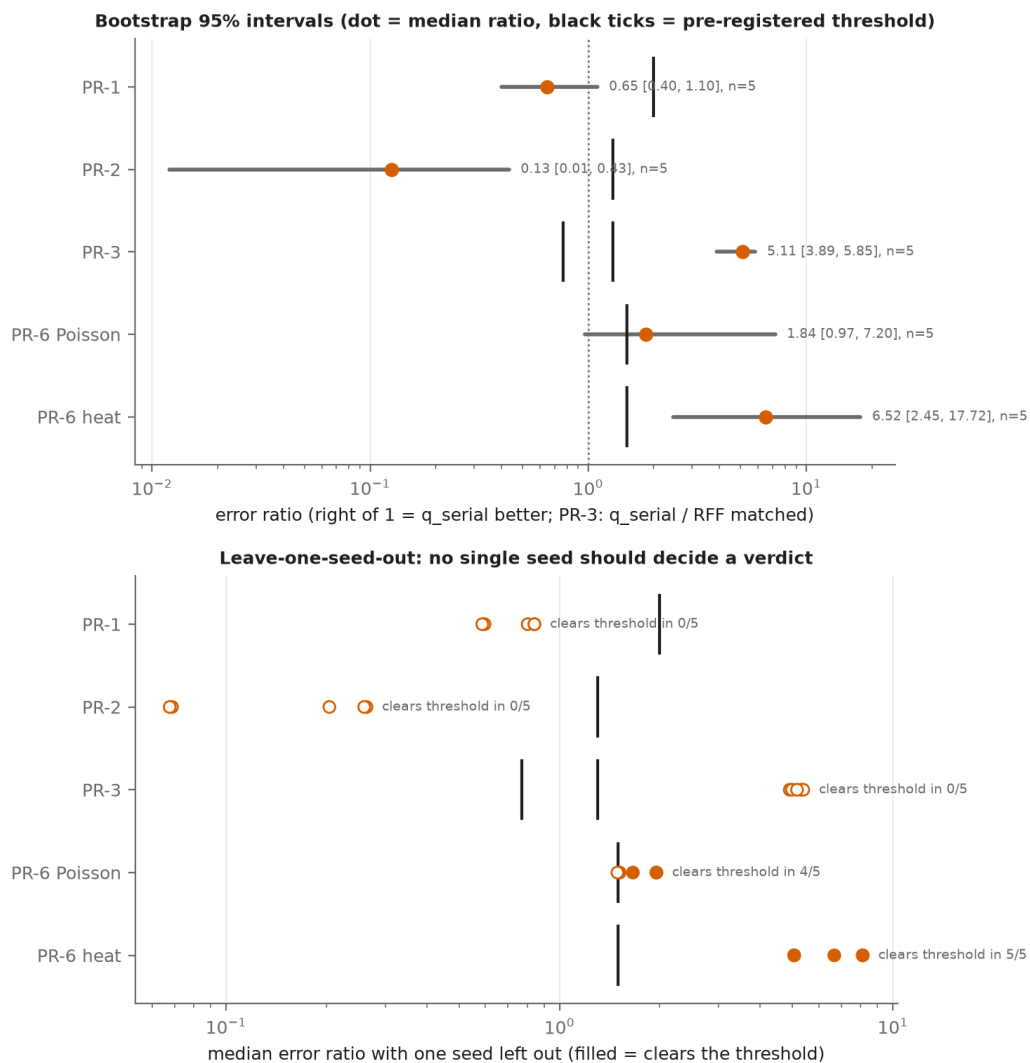
7 What the explainability instruments show



The circuits use very few directions in parameter space: NTK effective rank about 1 and Fisher dimension about 4, flat through training. The MLP's grows from 4 to 12 on Poisson and from 5 to 50 on heat. A size-matched classical model looks like the circuit, so this is mostly about size, not about being quantum.



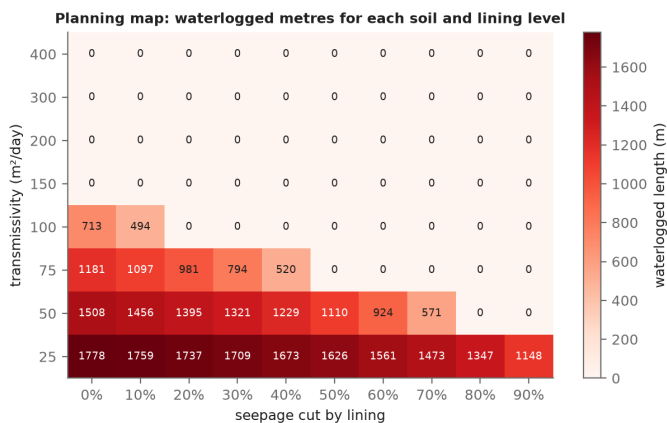
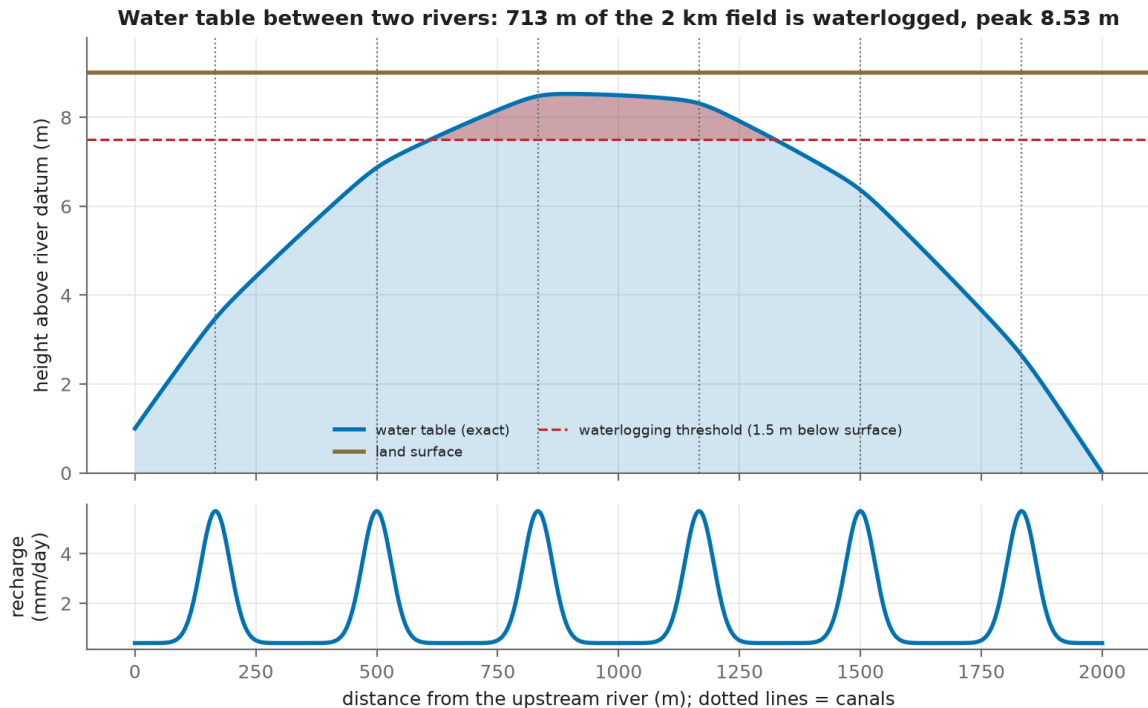
8 Are the verdicts robust?



The heat result's whole 95% interval clears its bar, and it clears it with any one seed removed. The Poisson losses miss their bars with any seed removed. Only the Poisson design-vs-random ratio moves across its bar (4 of 5), and that verdict is already inconclusive. Three other statistical tests agree with the pre-registered one.

9 A real problem: waterlogged farmland

Two rivers 2 km apart bound an aquifer under farmland irrigated by six leaking canals. Where the water table rises within 1.5 m of the surface, crop roots drown. This is the same equation as our Poisson problem, and the canal spacing tells SMCD the frequencies in advance.



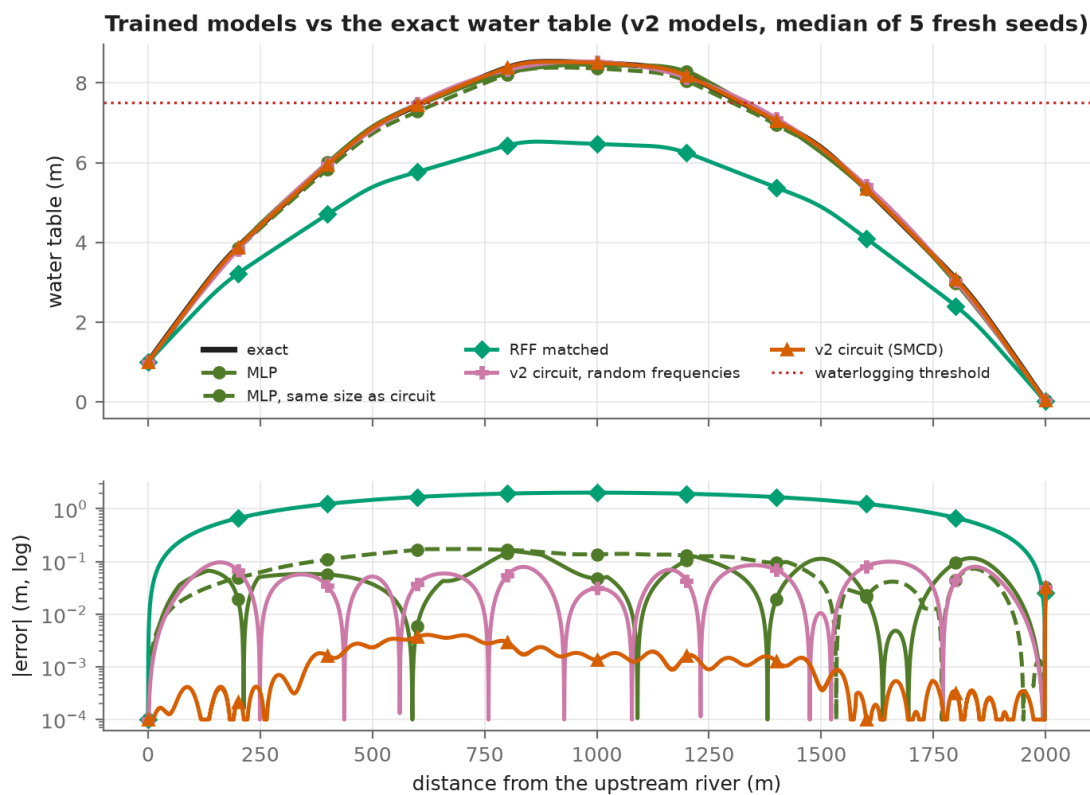
What the model tells a planner

- 713 m of the field is waterlogged to-day (peak 8.53 m).
- Lining the canals to cut seepage by 10% leaves 494 m; by 20%, none.
- On clay loam the same canals water-log 1778 m.

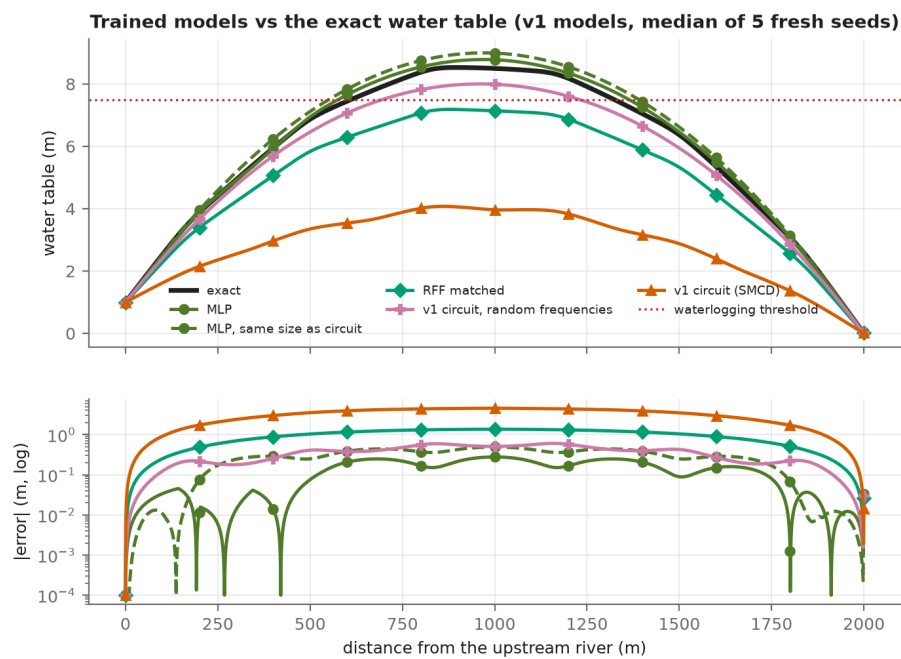
Can the trained models answer it?

We locked five predictions in git before training on fresh seeds, then scored them with a script, first for the original circuit and then for the improved v2 circuit (eight copies of the same design; see the next section for how it was built).

Prediction	original circuit	v2 circuit
G-1 error below 1%	no (53%)	yes (0.09%)
G-2 beats a same-size classical network	no	yes (18×)
G-3 beats the same circuit with random frequencies	no	yes (9.9×)
G-4 improvement lies on the designed frequencies	yes (99.4%)	yes (99.4%)
G-5 waterlogged length within 5% of the truth	no (0 m)	yes (714 m vs 713 m)
Held	1 of 5	5 of 5

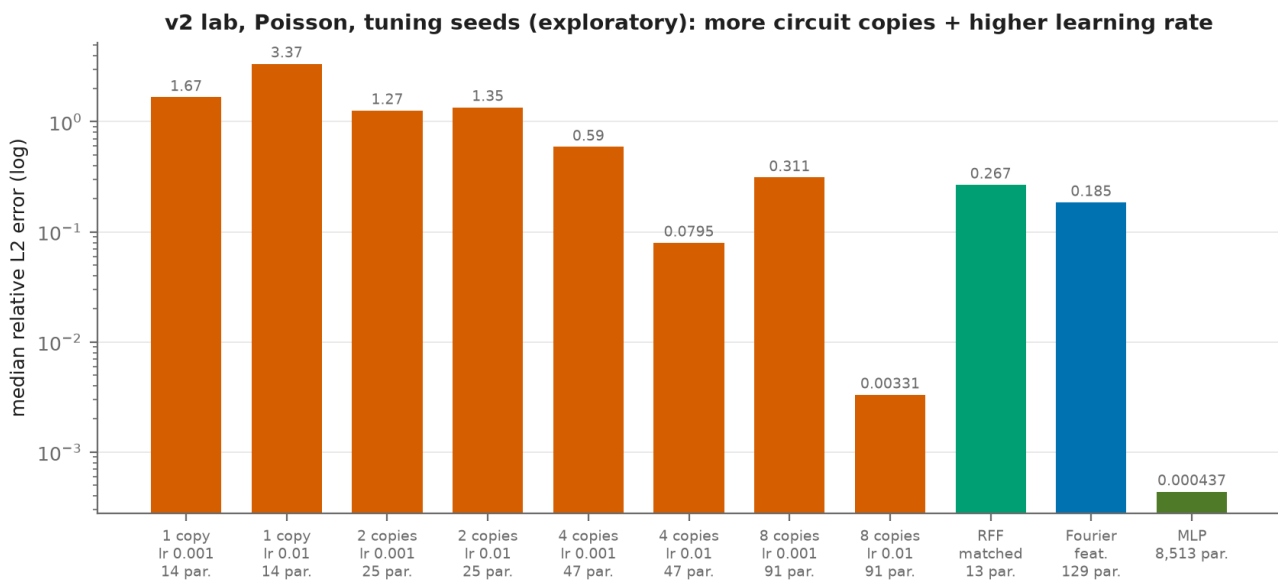


The v2 circuit tracks the true water table to within a few millimetres across the whole field and gets the waterlogged length right to 1 m. The original circuit (below) missed it entirely. The classical model handed the same frequencies fails here too.



10 Next: an improved quantum model (v2)

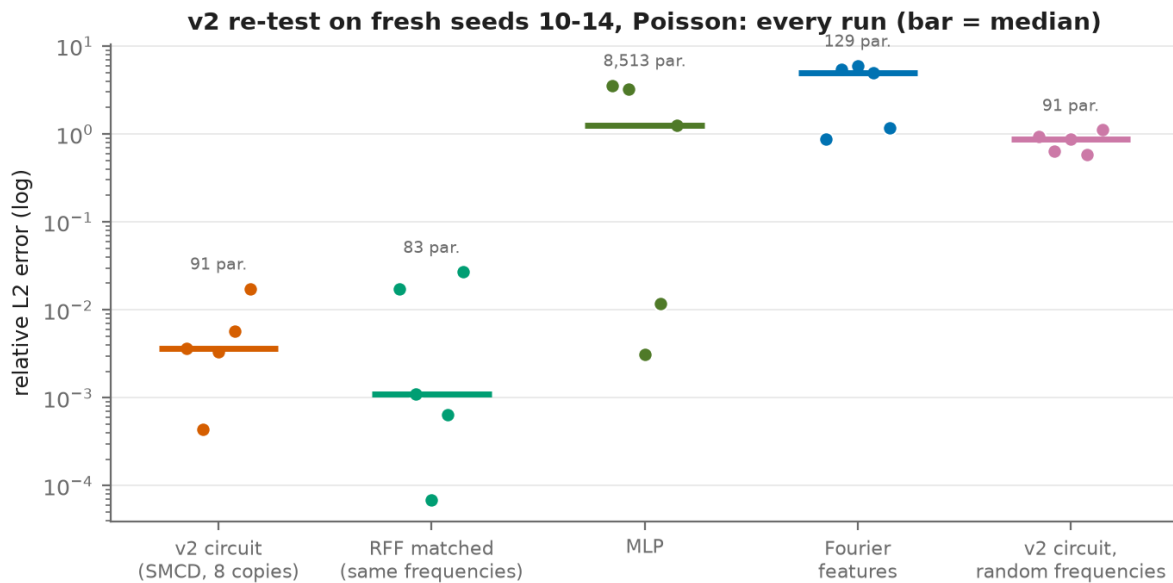
The diagnosis points to the fix: give the output layer independent handles on the wave amplitudes. v2 uses several copies of the designed circuit (same frequencies, their own angles), mixed by a linear head. We tuned it on the old seeds and locked new predictions in git before testing it on fresh ones.



On the tuning seeds, 8 copies (91 parameters) cut the Poisson error from 1.67 to 0.003: 56-81× better than the classical models with 13-129 parameters, but still behind the 8,513-parameter MLP. Tuning results are optimistic by construction; the locked re-test decides.

The locked re-test on fresh seeds

Eight of v1's claims, word for word, were re-tested on seeds 10-14 with the v2 circuit, after the predictions were pushed to GitHub. **5 of 8 held.**



- **Held:** beats Fourier features; beats the same circuit with random frequencies (236×); its improvement lands on the designed frequencies; the NTK is flatter inside Ω than outside; the encoder stays put.
- **Did not hold:** a classical model with the *same* frequencies and nearly the same size (83 parameters) is still 3× more accurate; the circuit does not leave the lazy-training regime earlier.
- **Inconclusive:** against the MLP. The MLP at its tuned learning rate failed on three of five fresh seeds and did well on two, so the seeds do not agree.
- **Caveat:** the classical learning rates chosen in tuning did not carry over to new seeds (the MLP and Fourier features were unstable), while every v2 circuit run landed below 2% error.

Summary

- **A design rule that works as specified.** SMCD builds circuits whose frequencies are exactly the ones designed, and those frequencies beat random ones on every seed of the heat equation.
- **The original circuit did not beat the classical baselines**, and we traced why: its wave amplitudes are coupled, so it trades the easy frequency for the hard one. Fixing that (eight copies) is what made the improved circuit work.
- **Honest by construction.** Predictions were locked before the runs, a script decided every verdict, and the full protocol was completed after the challenge without moving a threshold.
- **A real use, honestly scored.** The groundwater model answers a planning question (lining canals by 20% dries the field). The original circuit failed on it (1 of 5 locked predictions); the improved circuit met all 5, beating a same-size classical network 18× and the random-frequency circuit 9.9×.
- **Still open.** On Poisson a classical model given the same frequencies stays 3× ahead, and everything here runs on a simulator, not quantum hardware.

Links

Code, data, every run and every figure: https://github.com/Mannava-Daasaradhi/QAPINN_Spectra (branch v2 for this report). Full technical paper: `paper/main.pdf` in the repository.